MTH207 2020 S2 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

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* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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MTH207

Timed Online Assignment – July Semester 2020

Linear Algebra

Tuesday, 10 November 2020 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, SUSS PI No., and Your Name. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Let V=$\left\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R}\right\}$ be a set.

Let S= $\left\{\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ be a set of four vectors in V.

(a) Determine if S is a linearly independent set.

(5 marks)

(b) Given x, y, z, w are real numbers, find a, b, c, d such that

$$a\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} + d\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

(6 marks)

(c) Find a basis for V and state the dimension of V.

(3 marks)

(d) Does $\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ belong to Span$\left\{\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$?

(4 marks)

(e) State the dimension of the subspace Span$\left\{\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ of V.

(2 marks)

Question 2

(a) Compute the row reduced echelon form of the following matrix.

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{bmatrix}$$

(8 marks)

(b) Hence, or otherwise, solve the following system of linear equations:

$$\begin{cases} x - y + 2z = -3 \\ 4x + 4y - 2z = 1 \\ -x + y - 2z = 3 \end{cases}.$$

(3 marks)

(c) Given a system of linear equations in 3 variables

$$\begin{cases} x + 3y - 2z = 2a + c \\ \qquad\quad y + 2z = 6 \\ (c-3)z = ac + 3a \end{cases}.$$

Solve for which values of a and c does the system have

(i) exactly one solution.

(3 marks)

(ii) infinitely many solutions.

(3 marks)

(iii) no solutions.

(3 marks)

Question 3

(a) Calculate all possible values of b for which $\frac{1}{343}\begin{vmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{vmatrix} = 1$. Explain with the help of determinant if $\boldsymbol{A} = \frac{1}{7}\begin{bmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix}$ is invertible with these values of b.

(8 marks)

(b) For the positive value of b obtained in part (a), show that $\boldsymbol{A}^T\boldsymbol{A} = kI$ and solve for the value of k. Is $\boldsymbol{A}$ orthogonal?

(8 marks)

(c) Hence, deduce the matrix inverse of $\begin{bmatrix} 2 & b_1 & 6 \\ b_1 & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix}$, where b_1 is the positive value of b obtained in part (a).

(4 marks)

Question 4

Let $T : \mathbb{R}^P \to \mathbb{R}^Q$ is defined by $T(e_1) = e_2$, $T(e_2) = e_3$, $T(e_3) = 8e_1 - 6e_2 + 12e_3$, where $\boldsymbol{e_1} = \begin{bmatrix}1\\0\\0\end{bmatrix}$, $\boldsymbol{e_2} = \begin{bmatrix}0\\1\\0\end{bmatrix}$, $\boldsymbol{e_3} = \begin{bmatrix}0\\0\\1\end{bmatrix}$ are the standard basis of $\mathbb{R}^P$. You may assume T is a linear transformation.

(a) Write down the values of P and the minimal value of Q and demonstrate the matrix representation $\boldsymbol{A}$ of T with respect to standard basis in domain and codomain.

(6 marks)

(b) Compute bases for $Col\ (\boldsymbol{A})$, the column space of $\boldsymbol{A}$ and the dimension of $Nul\ (\boldsymbol{A})$, the null-space of $\boldsymbol{A}$.

(6 marks)

(c) Given a new basis $X = \{x_1, x_2, x_3\}$ of $\mathbb{R}^P$, where

$$\boldsymbol{x_1} = e_1, \quad \boldsymbol{x_2} = (\boldsymbol{A} - 2\boldsymbol{I})\, e_1, \quad \boldsymbol{x_3} = (\boldsymbol{A} - 2\boldsymbol{I})^2\, e_1$$

where $\boldsymbol{I}$ is the 3×3 identity matrix. Compute the vectors $\boldsymbol{x_1}, \boldsymbol{x_2}, \boldsymbol{x_3}$.

(4 marks)

(d) Find the matrix of T with respect to new basis $\{x_1, x_2, x_3\}$ in $\mathbb{R}^P$ and $\mathbb{R}^Q$.

(4 marks)

Question 5

(a) Let the matrix $\boldsymbol{A} = \begin{bmatrix}3 & 0 & 0\\-3 & 4 & 9\\0 & 0 & 3\end{bmatrix}$ and $\boldsymbol{e_1} = \begin{bmatrix}3\\0\\1\end{bmatrix}$ and $\boldsymbol{e_2} = \begin{bmatrix}1\\3\\0\end{bmatrix}$ are eigenvectors of $\boldsymbol{A}$ in $\mathbb{R}^3$.

(i) Calculate the remaining eigenvalue and corresponding eigenvector of $\boldsymbol{A}$.

(5 marks)

(ii) Solve for all possible invertible matrices $\mathbf{P}$ and the positive integer k such that

$$\mathbf{P}^{-1} \boldsymbol{A}^k \mathbf{P} = \begin{bmatrix}16 & 0 & 0\\0 & 9 & 0\\0 & 0 & 9\end{bmatrix}.$$

(5 marks)

(b) Apply the Gram-Schmidt process to obtain an orthogonal basis for the subspace of $\mathbb{R}^4$ spanned by

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

(10 marks)

----- END OF PAPER -----

1(a) Determine if set S is linearly independant

let $a\begin{bmatrix}1&1\\0&0\end{bmatrix}+b\begin{bmatrix}1&-1\\0&0\end{bmatrix}+c\begin{bmatrix}1&0\\-1&0\end{bmatrix}+d\begin{bmatrix}1&0\\0&1\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$

$\Leftrightarrow a\begin{bmatrix}1\\0\end{bmatrix}+b\begin{bmatrix}1\\0\end{bmatrix}+c\begin{bmatrix}1\\-1\end{bmatrix}+d\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$

$a\begin{bmatrix}1\\0\end{bmatrix}+b\begin{bmatrix}-1\\0\end{bmatrix}+c\begin{bmatrix}0\\0\end{bmatrix}+d\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$

$\Leftrightarrow \begin{bmatrix}1&1&1&1\\0&0&-1&0\\1&-1&0&0\\0&0&0&1\end{bmatrix}\begin{bmatrix}a&b&c&d\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$

$$\left[\begin{array}{cccc|c}1&1&1&1&0\\0&0&-1&0&0\\1&-1&0&0&0\\0&0&0&1&0\end{array}\right]\xrightarrow[r_3=-r_2]{r_2=\frac{1}{2}(r_1-r_3)}\left[\begin{array}{cccc|c}1&1&1&1&0\\0&1&\frac{1}{2}&\frac{1}{2}&0\\0&0&1&0&0\\0&0&0&1&0\end{array}\right]$$

$$\xrightarrow[r_1=r_1-r_2-\frac{1}{2}r_3-\frac{1}{2}r_4]{r_2=r_2-\frac{1}{2}r_3-\frac{1}{2}r_4}\left[\begin{array}{cccc|c}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\end{array}\right]$$

$\therefore a=b=c=d=0$ is the unique solution

$\therefore$ set S is linearly independant

1(b)

$$a\begin{bmatrix}1&1\\0&0\end{bmatrix}+b\begin{bmatrix}1&-1\\0&0\end{bmatrix}+c\begin{bmatrix}1&0\\-1&0\end{bmatrix}+d\begin{bmatrix}1&0\\0&1\end{bmatrix}-\begin{bmatrix}x&y\\z&w\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$$

$$a\begin{bmatrix}1&1\\0&0\end{bmatrix}+b\begin{bmatrix}1&-1\\0&0\end{bmatrix}+c\begin{bmatrix}1&0\\-1&0\end{bmatrix}+d\begin{bmatrix}1&0\\0&1\end{bmatrix}=\begin{bmatrix}x&y\\z&w\end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix}1&1&1&1\\0&0&-1&0\\1&-1&0&0\\0&0&0&1\end{bmatrix}\begin{bmatrix}a\\b\\c\\d\end{bmatrix}=\begin{bmatrix}x\\z\\y\\w\end{bmatrix}$$

$$\left[\begin{array}{cccc|c}1&1&1&1&x\\0&0&-1&0&z\\1&-1&0&0&y\\0&0&0&1&w\end{array}\right]\xrightarrow{r_3=r_3-r_1}\left[\begin{array}{cccc|c}1&1&1&1&x\\0&0&-1&0&z\\0&-2&-1&-1&y-x\\0&0&0&1&w\end{array}\right]$$

$$\downarrow r_2,r_3=r_3,r_2$$

$$\left[\begin{array}{cccc|c}1&1&1&1&x\\0&1&\frac{1}{2}&\frac{1}{2}&\frac{y-x}{2}\\0&0&1&0&-z\\0&0&0&1&w\end{array}\right]\xleftarrow[r_3=-r_3]{r_2=-\frac{1}{2}r_2}\left[\begin{array}{cccc|c}1&1&1&1&x\\0&-2&-1&-1&y-x\\0&0&-1&0&z\\0&0&0&1&w\end{array}\right]$$

$$\downarrow r_2=r_2-\tfrac{1}{2}r_3-\tfrac{1}{2}r_4$$

$$\left[\begin{array}{cccc|c}1&1&1&1&x\\0&1&0&0&-\frac{w}{2}+\frac{x}{2}-\frac{y}{2}+\frac{z}{2}\\0&0&1&0&-z\\0&0&0&1&w\end{array}\right]\xrightarrow{r_1=r_1-r_2-r_3-r_4}\left[\begin{array}{cccc|c}1&0&0&0&-\frac{w}{2}+\frac{x}{2}+\frac{y}{2}-\frac{z}{2}\\0&1&0&0&-\frac{w}{2}+\frac{x}{2}-\frac{y}{2}+\frac{z}{2}\\0&0&1&0&-z\\0&0&0&1&w\end{array}\right]$$

$\therefore a = -\frac{w}{2}+\frac{x}{2}+\frac{y}{2}-\frac{z}{2}$ $\qquad$ $\therefore c = -z$

$\therefore b\ -\frac{w}{2}+\frac{x}{2}-\frac{y}{2}+\frac{z}{2}$ $\qquad$ $\therefore d = w$

(c)

$$\text{basis} = \left\{\begin{bmatrix}1&1\\0&0\end{bmatrix},\begin{bmatrix}1&-1\\0&0\end{bmatrix},\begin{bmatrix}1&0\\-1&0\end{bmatrix},\begin{bmatrix}1&0\\0&1\end{bmatrix}\right\}$$

The dimension of V is 4

1(d) let $a\begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}+b\begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}+c\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}=\begin{bmatrix}1 & 2\\-2 & 1\end{bmatrix}$

$$\Leftrightarrow \begin{bmatrix}1 & 1 & 1\\0 & -1 & 0\\-1 & 0 & 0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}a & b & c\end{bmatrix} = \begin{bmatrix}1\\-2\\2\\1\end{bmatrix}$$

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 1\\0 & -1 & 0 & -2\\-1 & 0 & 0 & 2\\0 & 0 & 1 & 1\end{array}\right] \xrightarrow[r_3=r_3+r_1+r_2]{r_2=-r_2} \left[\begin{array}{ccc|c}1 & 1 & 1 & 1\\0 & -1 & 0 & -2\\0 & 1 & 1 & 3\\0 & 0 & 1 & 1\end{array}\right]$$

$$\xrightarrow[r_3=r_3+r_2]{r_2=-r_2} \left[\begin{array}{ccc|c}1 & 1 & 1 & 1\\0 & 1 & 0 & 2\\0 & 0 & 1 & 1\\0 & 0 & 1 & 1\end{array}\right] \xrightarrow[r_4=r_4-r_3]{r_1=r_1-r_2-r_3} \left[\begin{array}{ccc|c}1 & 0 & 0 & -2\\0 & 1 & 0 & 2\\0 & 0 & 1 & 1\\0 & 0 & 0 & 0\end{array}\right]$$

$a=-2,\ b=2,\ c=1$

$$\begin{bmatrix}1 & 2\\-2 & 1\end{bmatrix} = -2\begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}+2\begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}+\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

$\therefore \begin{bmatrix}1 & 2\\-2 & 1\end{bmatrix}$ belongs to $\text{span}\left\{\begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\right\}$

(e) $\begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}$ is not a scalar multiple of $\begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

$\begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}$ is not a scalar multiple of $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

$\left\{\begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\-1 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\right\}$ is linearly independent

Hence, the dimension is 3

2(a)

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{bmatrix} \xrightarrow[r_3 = r_3 + 2r_1]{r_2 = r_2 - 4r_1} \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 8 & -10 & 13 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[r_3 = \frac{1}{8} r_2]{r_1 = r_1 + \frac{1}{8} r_2} \begin{bmatrix} 1 & 0 & 3/4 & -11/8 \\ 0 & 1 & -5/4 & 13/8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$x - y + 2z = -3$$
$$4x + 4y - 2z = 1$$
$$-x + y - 2z = 3 \Rightarrow -2x + 2y - 4z = -6$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 4 & 4 & -2 \\ -2 & 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 6 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 3/4 & -11/8 \\ 0 & 1 & -5/4 & 13/8 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -\frac{11}{8} - \frac{3}{4} z \\ \frac{13}{8} + \frac{5}{4} z \\ z \end{bmatrix} = \frac{1}{8} \begin{bmatrix} -11 \\ 13 \\ 0 \end{bmatrix} + \frac{z}{4} \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}$$

2(c)

		outcome
$c=3$	$a=0$	infinite solution
	$a\neq 0$	no solution
$c\neq 3$	$a=0$	unique solution
	$a\neq 0$	unique solution

case 1: $c=3,\ a=0$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & c-3 & a+3a \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 3 & -2 & 3 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\downarrow r_1 = r_1 - 3r_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & -15 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Infinite solutions:

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8z+15 \\ 6-2z \\ z \end{bmatrix}$$

case 2: $c=3,\ a\neq 0$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & c-3 & a+3a \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a-3 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 4a \end{array}\right]$$

$\therefore$ last row not consistent

$\therefore$ no solution exists for case 2

2(c)
$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & c-3 & ac+3a \end{array}\right]$$

$$\Big\downarrow\ r_3 = \frac{r_3}{c-3}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 1 & (ac+3a)/(c-3) \end{array}\right]$$

$$\Big\downarrow\ \begin{array}{l} r_1 = r_1 + 2r_3 \\ r_2 = r_2 - 2r_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2a+c+2(ac+3a)/(c-3) \\ 0 & 1 & 0 & 6-2(ac+3a)/(c-3) \\ 0 & 0 & 1 & (ac+3a)/(c-3) \end{array}\right]$$

$$\Big\downarrow\ r_1 = r_1 - 3r_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2a+c-18+8(ac+3a)/(c-3) \\ 0 & 1 & 0 & 6-2(ac+3a)/(c-3) \\ 0 & 0 & 1 & (ac+3a)/(c-3) \end{array}\right]$$

Case 3: $c \neq 3$, $a = 0$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & c-3 & ac+3a \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & c-18 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} c-18 \\ 6 \\ 0 \end{bmatrix}$$

$\therefore$ Case for unique solution

Case 4: $c \neq 3$, $a \neq 0$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 2a+c \\ 0 & 1 & 2 & 6 \\ 0 & 0 & c-3 & ac+3a \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2a+c-18+8(ac+3a)/(c-3) \\ 0 & 1 & 0 & 6-2(ac+3a)/(c-3) \\ 0 & 0 & 1 & (ac+3a)/(c-3) \end{array}\right]$$

$\therefore$ Case for unique solution

3(a) $A = \frac{1}{7}\begin{bmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1/7 & 0 & 0 \\ 0 & 1/7 & 0 \\ 0 & 0 & 1/7 \end{bmatrix}\begin{bmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix}$

$$\det A = \begin{vmatrix} 1/7 & 0 & 0 \\ 0 & 1/7 & 0 \\ 0 & 0 & 1/7 \end{vmatrix} \begin{vmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{vmatrix}$$

$$= \frac{1}{343}\begin{vmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{vmatrix}$$

since $\det A = 1$, then $\begin{vmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 6 & 2 & -3 \end{vmatrix} = 343$

$\downarrow r_3 = r_3 - 3r_1$

$$\begin{vmatrix} 2 & b & 6 \\ b & -6 & 2 \\ 0 & 2-3b & -21 \end{vmatrix} = 343$$

$$2\begin{vmatrix} -6 & 2 \\ 2-3b & -21 \end{vmatrix} - b\begin{vmatrix} b & 6 \\ 2-3b & -21 \end{vmatrix} - 343 = 0$$

$$3b^2 + 24b - 99 = 0$$

$$b^2 + 8b - 33 = 0$$

$$(b+11)(b-3) = 0$$

$$\therefore b = 3 \text{ or } -11$$

3(b) let $b=3$

$$A=\frac{1}{7}\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix} \qquad A^T=\frac{1}{7}\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}$$

$$A^TA=\frac{1}{7}\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}\times\frac{1}{7}\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}$$

$$=\frac{1}{49}\times\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}\times\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}$$

$$=\frac{1}{49}\begin{bmatrix}49&0&0\\0&49&0\\0&0&49\end{bmatrix}$$

$$=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$\therefore K=1$

Since $A^{-1}A=I$, then A is an orthogonal matrix

(c) let $B=\begin{bmatrix}2&b_1&6\\b_1&-6&2\\6&2&-3\end{bmatrix}$ where $b_1=3$

$A=\frac{1}{7}B$

$A^{-1}=A^T=\frac{1}{7}B^T$

$A^{-1}A=I$

$(\frac{1}{7}B^T)(\frac{1}{7}B)=I$

$(\frac{1}{49}B^T)B=I$

$\therefore B^{-1}=\frac{1}{49}B^T$

$$=\frac{1}{49}\begin{bmatrix}2&3&6\\3&-6&2\\6&2&-3\end{bmatrix}.$$

4(a) $P = 3$, $\min Q = 3$

let $T(v) = Av$ where A is the transformation matrix

$T(e_1) = e_2$, $T(e_2) = e_3$, $T(e_3) = 8e_1 - 6e_2 + 12e_3$

$$\Rightarrow A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 8 \\ 1 & 0 & -6 \\ 0 & 1 & 12 \end{bmatrix}$$

since $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

$$\therefore A = \begin{bmatrix} 0 & 0 & 8 \\ 1 & 0 & -6 \\ 0 & 1 & 12 \end{bmatrix}$$

(b) let $Ax = 0$

$$\left[\begin{array}{ccc|c} 0 & 0 & 8 & 0 \\ 1 & 0 & -6 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right] \xrightarrow{r_1, r_2, r_3 = r_2, r_3, r_1} \left[\begin{array}{ccc|c} 1 & 0 & -6 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 8 & 0 \end{array}\right]$$

$$\xrightarrow{r_2 = r_2 - \frac{1}{4} r_3,\ r_3 = \frac{1}{8} r_3} \left[\begin{array}{ccc|c} 1 & 0 & -6 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{r_1 = r_1 + 6r_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$\therefore$ The column vectors of A are linearly independent

$$\therefore \text{col}(A) = \text{span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 8 \\ -6 \\ 2 \end{bmatrix} \right\}$$

This tells us that A is invertible, so $\text{Rank}(A) = 3$

By the Rank Nullity Theorem

$$\text{Nullity of } A = \text{rank}(A) - \dim(\text{col}(A))$$
$$= 3 - 3$$
$$= 0$$

4(c) $x_1 = e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$x_2 = (A-2I)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} -2 & 0 & 8 \\ 1 & -2 & -6 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$x_3 = (A-2I)^2 e_1$$
$$= \begin{bmatrix} -2 & 0 & 8 \\ 1 & -2 & -6 \\ 0 & 1 & 0 \end{bmatrix}^2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} -2 & 0 & 8 \\ 1 & -2 & -6 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix}$$

(d) $T(x_1) = T(e_1) = e_2 = 2x_1 + x_2 + 0x_3$

$$T(x_2) = T(e_2 - 2e_1)$$
$$= T(e_2) - 2T(e_1)$$
$$= e_3 - 2e_2$$
$$= \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$
$$= 2x_2 + x_3$$

$$T(x_3) = T(4e_1 - 4e_2 + e_3)$$
$$= 4T(e_1) - 4T(e_2) + T(e_3)$$
$$= 4e_2 - 4e_3 + (8e_1 - 6e_2 + 12e_3)$$
$$= 8e_1 - 2e_2 + 8e_3$$
$$= \begin{bmatrix} 8 \\ -2 \\ 8 \end{bmatrix} = 8\begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix} + 30\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + 36\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= 36x_1 + 30x_2 + 8x_3$$

$$\therefore C = \begin{bmatrix} 2 & 0 & 36 \\ 1 & 1 & 30 \\ 0 & 1 & 8 \end{bmatrix}$$

5(a) i. $Ax = \lambda x$

$(A-\lambda I)x = \underset{\sim}{0}$

let $\det(A-\lambda I) = 0$

$$\begin{vmatrix} 3-\lambda & 0 & 0 \\ -3 & 4-\lambda & 9 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)\begin{vmatrix} 4-\lambda & 9 \\ 0 & 3-\lambda \end{vmatrix} = 0$$

$(3-\lambda)((4-\lambda)(3-\lambda) - 0) = 0$

$(4-\lambda)(3-\lambda)^2 = 0$

$\therefore \lambda = 3$ or 4

Since we are looking for the eigenvalue of single algebraic multiplicity, remaining eigenvalue $\lambda = 4$

$$\left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ -3 & 0 & 9 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right] \xrightarrow{r_2 = r_2 - 3r_1} \left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$$

$r_1 \leftrightarrow r_3$

$r_2 = \frac{1}{9} r_2$

$r_3 = r_3 + \frac{1}{9} r_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

The corresponding eigenvector $= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

ii. $A = PDP^{-1}$

where $D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 3 & 1 \\ 1 & 0 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

$A^k = (PDP^{-1})(PDP^{-1}) \dots (PDP^{-1}) = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)\dots DP^{-1}$

$= PD^kP^{-1}$

$P^{-1}A^kP = P^{-1}(PD^kP^{-1})P$

$= (P^{-1}P)D^k(P^{-1}P)$

$= D^k$

$$\begin{bmatrix} 4^k & 0 & 0 \\ 0 & 3^k & 0 \\ 0 & 0 & 3^k \end{bmatrix} = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$\therefore k = 2$

5(c) $x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 2 \end{bmatrix}$ $x_3 = \begin{bmatrix} 0 \\ 1 \\ 4 \\ 1 \end{bmatrix}$

subspace = span $\{x_1, x_2, x_3\}$

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \frac{1}{7} \begin{bmatrix} 1 \\ -6 \\ 2 \\ 1 \end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 = \frac{1}{2} \begin{bmatrix} -3 \\ 0 \\ 2 \\ -1 \end{bmatrix}$$

$\therefore$ orthogonal basis $= \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -6 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ -1 \end{bmatrix} \right\}$